# Prelims Probability

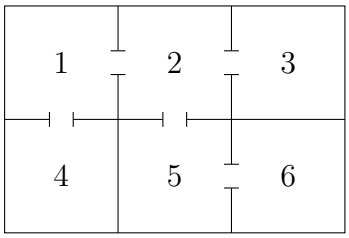
## Sheet 5 — MT26

1. A bug jumps around the vertices of a triangle. At every jump, it moves from its current position to either of the other two vertices with probability 1/2 each (independently of how it arrived at its current position). The bug starts at vertex 1. Let $p_n$ be the probabilility that it is at vertex 1 after $n$ jumps.

   (a) Find the value of $p_n$ for each $n$. [*Hint: find an appropriate first-order linear recurrence relation.*]

   (b) What happens to $p_n$ as $n \to \infty$?

2. The diagram below shows the floor plan of a house with six rooms: in room 1 is a mouse which will change rooms every minute, first moving at $t = 1$ and choosing a door to an adjoining room at random. In room 6 is a sleeping but hungry cat which will instantly wake if the mouse should enter. How long on average can we expect the mouse to survive?




3. (**Gambler's ruin, symmetric case.**) A gambler starts a game with a bankroll of $£n$ where $n \in \{1, 2, \ldots, M-1\}$. At each step of the game, he wins $£1$ with probability 1/2 and loses $£1$ with probability 1/2, independently for different steps. The game ends when the gambler's bankroll reaches $£0$ or $£M$.

   In lectures we saw that the probability the gambler finishes with $£M$ is $n/M$.

   (a) What is the expected amount of money that the gambler has at the end of the game?

   (b) Suppose we know that the gambler ends the game with $£M$. What is the conditional probability that he won $£1$ on the first step?

   (c) Let $e_n$ be the expected length of the game. Find $e_n$ for each $n$. For which $n$ is $e_n$ largest?


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4. (a) Suppose that $X$ has a geometric distribution with parameter $p$. Show that the probability generating function of $X$ is

$$G_X(s) = \frac{ps}{1-(1-p)s}, \qquad \text{for } |s| < \frac{1}{1-p}.$$

   (b) Use this to calculate the mean and variance of $X$.

5. (a) A fair coin is tossed $n$ times. Let $r_n$ be the probability that the sequence of tosses never has a head followed by a head. Show that

$$r_n = \frac{1}{2}r_{n-1} + \frac{1}{4}r_{n-2}, \quad n \geq 2.$$

   Find $r_n$ using the conditions $r_0 = r_1 = 1$. Check that the value you get for $r_2$ is correct.

   (b) Let $X$ be the number of coin tosses needed until you first get two heads in a row. (Note that $X \geq 2$.) Find the probability mass function of $X$.

   (c) Find the probability generating function of $X$. Use this to calculate the mean of $X$. *(You may wish to check that your answer agrees with what you got for Question 6 on Problem Sheet 3!)*

   (d) Let $Y$ be the number of coin tosses needed until you first see a tail followed by a head. On any two particular coin tosses, the probability of seeing the pattern TH is $1/4$, the same as the probability of seeing the pattern HH. Therefore $\mathbb{P}(Y = 2) = \mathbb{P}(X = 2) = 1/4$. Find $\mathbb{P}(Y > n)$ for $n \geq 1$ and compare it to $\mathbb{P}(X > n)$. Is your answer surprising?

6. (*Optional. If you liked the coupon collector problem on Problem Sheet 3, you may enjoy this question too!*)
   Consider a symmetric random walk on a cycle with $N$ sites, labelled $0, 1, 2, \ldots, N-1$. A particle starts at site 0, and at each step it jumps from its current site $i$ to one of its two neighbours $i+1 \mod N$ and $i-1 \mod N$ with equal probability (independently of how it arrived at its current position).

   (a) Find the expected number of steps until every site has been visited. [*Hint: just after a new site has been visited, what does the set of visited sites look like? The value $e_1 = M - 1$ from Question 3(c) may be useful!*]

   (b) For each $k = 1, \ldots, N-1$, what is the probability that $k$ is the last site to be visited? [*Hint: before visiting site $k$, the walk must visit either site $k-1$ or site $k+1$. What needs to happen from that point onwards?*]


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